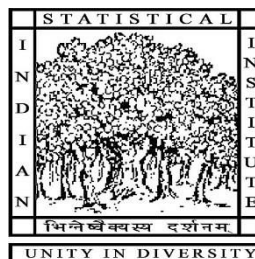


Ramakrishna Mission Residential College
Narendrapur, Kolkata-700103
An Autonomous College affiliated to the University of Calcutta



INTERNSHIP REPORT 2025

On

[Estimation of Parameter of Ratios of two Dependent Normal Distributions]

Submitted by

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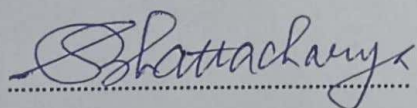
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Name of the Internship Supervisor: *Prof. Sabyasachi Bhattacharya*

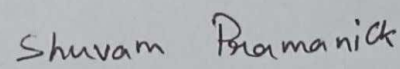
Affiliation of the Internship Supervisor: *Professor of A.E.R.U, ISI, Kolkata*

Remarks on the Internship report:

This is to acknowledge that **Shuvam Pramanick** attended our lab regularly, demonstrated strong theoretical problem-solving skills, and effectively analysed results using appropriate software tools. He successfully completed the assigned work with consistent and commendable performance throughout.

 23.07.25.

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Declaration

We hereby declare that the project titled “*Estimation of Parameter of Ratios of two Dependent Normal Distributions* ” has been carried out by us as part of our summer internship under the guidance of Prof. Sabyasachi Bhattacharya at the Indian Statistical Institute, Kolkata.

This work is the result of our joint efforts and has not been submitted elsewhere for any academic or professional purpose. All simulations, analyses, and results presented in this report were conducted using R programming, and appropriate references have been cited wherever necessary.

We have maintained academic integrity and followed ethical research practices throughout the course of this project.

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Shuvam Pramanick

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Introduction

Statistical estimators are formal functions of sample data designed to infer unknown population parameters, such as mean, variance, or correlation. Their evaluation is based on key properties including unbiasedness, consistency, efficiency, and robustness. The performance and suitability of an estimator are inherently tied to the underlying probability distribution of the data. In well behaved parametric models like the bivariate normal distribution, classical estimators such as the sample mean, sample variance, and sample correlation coefficient are widely used due to their favorable theoretical properties, often being unbiased, consistent, and asymptotically efficient. However, for heavy-tailed distributions such as the Cauchy, which lacks finite moments, these conventional estimators become unreliable or undefined. In such cases, alternative strategies, including robust estimators (e.g., the median) or distribution-specific methods like maximum likelihood estimation, are necessary. This highlights a central principle in statistical inference: The choice of estimator must be guided by the distributional characteristics of the data to ensure valid and meaningful estimation.

The estimation of the correlation parameter ρ in a bivariate setting is fundamental in statistics, as it quantifies the linear dependence between two jointly distributed random variables. As mentioned earlier when (X, Y) follows a bivariate normal distribution with parameters $(0, 0, 1, 1, \rho)$, the maximum likelihood estimation (MLE) method is a natural and efficient tool to estimate ρ using observed data.

Interestingly, under this setting, the ratio $Z = X/Y$ follows a Cauchy distribution with location parameter ρ and scale parameter $\sqrt{1 - \rho^2}$. This property provides an alternative perspective on the estimation of ρ , allowing us to explore estimation techniques under both the bivariate normal framework and the Cauchy distribution induced by the ratio.

Objectives

The objectives of this study are:

1. To explore the distribution of the ratio $Z = X/Y$ under the bivariate normal setting:
 - Prove that Z follows a Cauchy distribution with parameters $(\rho, \sqrt{1 - \rho^2})$

- Understand the heavy-tailed nature of the induced distribution from the perspective of correlation
2. To examine the estimation of ρ using the induced Cauchy distribution:
 - Apply MLE to derive an estimate for ρ using the observed ratios $Z_i = X_i/Y_i$
 - Apply order statistics to derive an estimate for ρ using the observed ratios $Z_i = X_i/Y_i$
 3. To compare the estimation of ρ under both frameworks:
 - Discuss consistency, efficiency, and practical considerations of using MLE under the bivariate normal distribution versus the Cauchy distribution induced by the ratio
 - Provide insights into when each method is preferable in practice

Work Done

Mathematical Derivation:

A Bivariate Normal distribution describes the joint probability distribution of two random variables that are both normally distributed. It is characterized by five parameters: the means and standard deviations of the two variables and their correlation coefficient. Here, X and Y are two random variables following a bivariate normal distribution with parameters $(0, 0, 1, 1, \rho)$

i.e. $(X, Y) \sim \mathcal{N}_2(0, 0, 1, 1, \rho)$

This means:

$$\mathbb{E}[X] = \mathbb{E}[Y] = 0, \quad \text{Var}(X) = \text{Var}(Y) = 1, \quad \text{Cov}(X, Y) = \rho$$

The joint PDF is:

$$f(x, y) = \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}(x^2 - 2\rho xy + y^2)\right)$$

To find the distribution of:

$$Z = \frac{X}{Y}$$

Computation

Let:

$$U = X, \quad V = Y$$

We want the distribution of:

$$Z = \frac{U}{V}$$

and let:

$$W = V$$

The Jacobian of the transformation from (U, V) to (Z, W) is:

$$J = \begin{vmatrix} \frac{\partial U}{\partial Z} & \frac{\partial U}{\partial W} \\ \frac{\partial V}{\partial Z} & \frac{\partial V}{\partial W} \end{vmatrix} = \begin{vmatrix} W & Z \\ 0 & 1 \end{vmatrix} = W$$

so the absolute value is $|W|$.

Joint PDF of (Z, W)

Since:

$$U = ZW, \quad V = W$$

The joint PDF becomes:

$$\begin{aligned} f_{Z,W}(z, w) &= f_{U,V}(zw, w) \cdot |w| \\ &= \frac{|w|}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} [(zw)^2 - 2\rho(zw)(w) + w^2]\right) \\ &= \frac{|w|}{2\pi\sqrt{1-\rho^2}} \exp\left(-\frac{w^2}{2(1-\rho^2)} [z^2 - 2\rho z + 1]\right) \end{aligned}$$

Marginal PDF of Z

To find the PDF of Z , integrate out W :

$$f_Z(z) = \int_{-\infty}^{\infty} f_{Z,W}(z, w) dw$$

Using symmetry and the known integral:

$$\int_0^{\infty} w \exp(-aw^2) dw = \frac{1}{2a}$$

where:

$$a = \frac{z^2 - 2\rho z + 1}{2(1 - \rho^2)}$$

We get:

$$f_Z(z) = \frac{1}{\pi} \cdot \frac{\sqrt{1 - \rho^2}}{(z - \rho)^2 + (1 - \rho^2)}$$

Deduction

$$Z = \frac{X}{Y} \sim \text{Cauchy}(\rho, \sqrt{1 - \rho^2})$$

with PDF:

$$f_Z(z) = \frac{1}{\pi} \cdot \frac{\gamma}{(z - x_0)^2 + \gamma^2}$$

where:

$$x_0 = \rho, \quad \gamma = \sqrt{1 - \rho^2}$$

This completes the proof that the ratio of jointly normal variables with unit variances and correlation ρ follows a Cauchy distribution with location ρ and scale $\sqrt{1 - \rho^2}$.

Implementation In R

We perform a comprehensive study aimed at evaluating multiple estimators of the unknown correlation parameter ρ using the ratio $Z = X/Y$, where (X, Y) are jointly drawn from a bivariate normal distribution $(0, 0, 1, 1, \rho)$. The data is simulated from

bivariate normal distribution with a fixed true correlation of $\rho = 0.5$. Since the ratio of two normally distributed variables exhibits heavy-tailed behavior, often resembling a Cauchy distribution, this setup serves to test the robustness and reliability of different estimation methods under non-Gaussian conditions. Here we estimate ρ using induced Cauchy distribution.

The simulation is carried out over 40 iterations with sample sizes increasing from 25 to 1000 in increments of 25. In each iteration, a dataset is generated, and the ratio $Z = X/Y$ is computed.

Estimators Used

- **Estimator 1:** Median of Z following Cauchy distribution with parameters $(\rho, \sqrt{1 - \rho^2})$.

$$estimator_1 = \text{median}(Z)$$

- **Estimator 2:** *Estimator₂ is derived from the maximum likelihood estimation (MLE) of the location parameter of the induced Cauchy distribution(i.e. from the distribution of Z) with location parameter ρ . It is estimated using the `cauchy.mle()` function from the `cauchypca` package.*
 $estimator_2 = \text{location MLE of } Z$

- **Estimator 3:** *Sample Correlation of Bivariate Normal distribution with parameters $(0,0,1,1,\rho)$ is taken as estimator 3.*

$$estimator_3 = r$$

where, r =Pearson's Product Moment Correlation Coefficient between X and Y

- **Estimator 4:** *An estimator derived from the maximum likelihood estimation (MLE) of the scale parameter of the induced Cauchy distribution(i.e. from the distribution of Z) . (SC), where the scale parameter is given by $\sqrt{1 - \rho^2}$. It is estimated using the `cauchy.mle()` function from the `cauchypca` package.*
 $SC = \text{MLE of Cauchy distribution}$

$$estimator_4 = \sqrt{1 - (SC)^2}$$

- **Estimator 5:** *An estimator derived from the quartile deviation of a random variable Z following a Cauchy distribution (QD) with location parameter ρ and scale parameter $\sqrt{1 - \rho^2}$.*
 $QD = [Q_3 - Q_1]/2$

$$estimator_5 = \sqrt{1 - (QD)^2}$$

Tabulation Of The Estimation

Table 1: Estimated Values of Different Estimators Across Sample Sizes

sample_size	estimator_1	estimator_2	estimator_3	estimator_4	estimator_5
25	0.00131	0.0592	0.59772	0.92644	0.96195
50	0.43699	0.55038	0.37256	0.59291	NA
75	0.70828	0.6943	0.61696	0.64493	0.74473
100	0.67711	0.6103	0.47938	0.58783	0.70437
125	0.662	0.52772	0.43606	0.48361	0.58652
150	0.47229	0.46221	0.43923	0.55358	0.58553
175	0.61886	0.55987	0.46453	0.05551	NA
200	0.56318	0.56796	0.51216	0.49644	0.36922
225	0.52457	0.5188	0.50692	0.57373	0.54765
250	0.4016	0.4129	0.4731	0.44097	0.5469
275	0.3954	0.42943	0.48637	0.71377	0.63359
300	0.41437	0.43717	0.47862	0.51665	0.57335
325	0.45399	0.45054	0.47369	0.53027	0.54664
350	0.4254	0.50011	0.4715	NA	NA
375	0.49245	0.50522	0.4925	0.43103	0.44148
400	0.60713	0.55152	0.5198	0.43837	0.46528
425	0.50726	0.48792	0.51049	0.527	0.54763
450	0.59035	0.52333	0.52389	0.48686	0.53139
475	0.54127	0.51083	0.51794	0.4839	0.50334
500	0.61108	0.55858	0.55646	0.62341	0.59211
525	0.61574	0.58626	0.55505	0.51672	0.48919
550	0.65108	0.61942	0.55135	0.48094	0.48637
575	0.53637	0.53553	0.55503	0.65412	0.65451
600	0.55314	0.54208	0.54444	0.61375	0.66406

(continued from previous page)

sample_size	estimator_1	estimator_2	estimator_3	estimator_4	estimator_5
625	0.5251	0.50792	0.53941	0.66831	0.6667
650	0.5056	0.49096	0.53497	0.62731	0.65159
675	0.58029	0.55811	0.53003	0.53469	0.49539
700	0.55767	0.54604	0.53159	0.559	0.56376
725	0.52697	0.52476	0.53836	0.5732	0.59304
750	0.51193	0.53786	0.53978	0.58398	0.58813
775	0.46595	0.53083	0.53858	0.49115	0.51155
800	0.57694	0.53231	0.53378	0.562	0.5436
825	0.5348	0.52614	0.528	0.52946	0.52277
850	0.53232	0.53945	0.52445	0.4725	0.46894
875	0.44715	0.47924	0.51053	0.61354	0.59914
900	0.49188	0.46993	0.51032	0.55959	0.59404
925	0.47849	0.49923	0.50358	0.49902	0.50848
950	0.5115	0.50728	0.50853	0.48348	0.54291
975	0.5308	0.51505	0.51468	0.46333	0.50255
1000	0.44831	0.45268	0.51275	0.61458	0.62773

The above table presents the estimated values of the correlation parameter obtained using various statistical estimators across different sample sizes. It is evident that for small sample sizes, the estimates of ρ exhibit substantial deviation from the true population value. This variability can be attributed to the high sampling variability and the sensitivity of some estimators to small samples, particularly in the presence of heavy-tailed or non-normal data.

However, as the sample size increases, the accuracy and reliability of all the estimators improve noticeably. This trend aligns with the theoretical expectation that most consistent estimators converge to the true parameter value as the sample size grows, thereby enhancing their performance in terms of bias, variance, and overall reliability.

- **Note:** Some NA values are observed in the table for Estimator 4 and Estimator 5 at certain sample sizes. This occurs because these estimators are constructed based on specific transformations of the estimator of scale parameter of the induced Cauchy distribution. However, due to the heavy-tailed nature of the Cauchy distribution, the estimated scale parameter can occasionally exceed one. In such cases, the resulting computation becomes undefined, leading to

NA values in the output.

Visualisation Of Result

To evaluate and compare the performance of the various estimators of ρ from the induced Cauchy distribution using the Pearson product moment correlation estimator a visual tool is used. In this section we plot the estimated values of different estimators along increasing sample sizes, i.e., the result of Table 1 using the line-plot function in R.

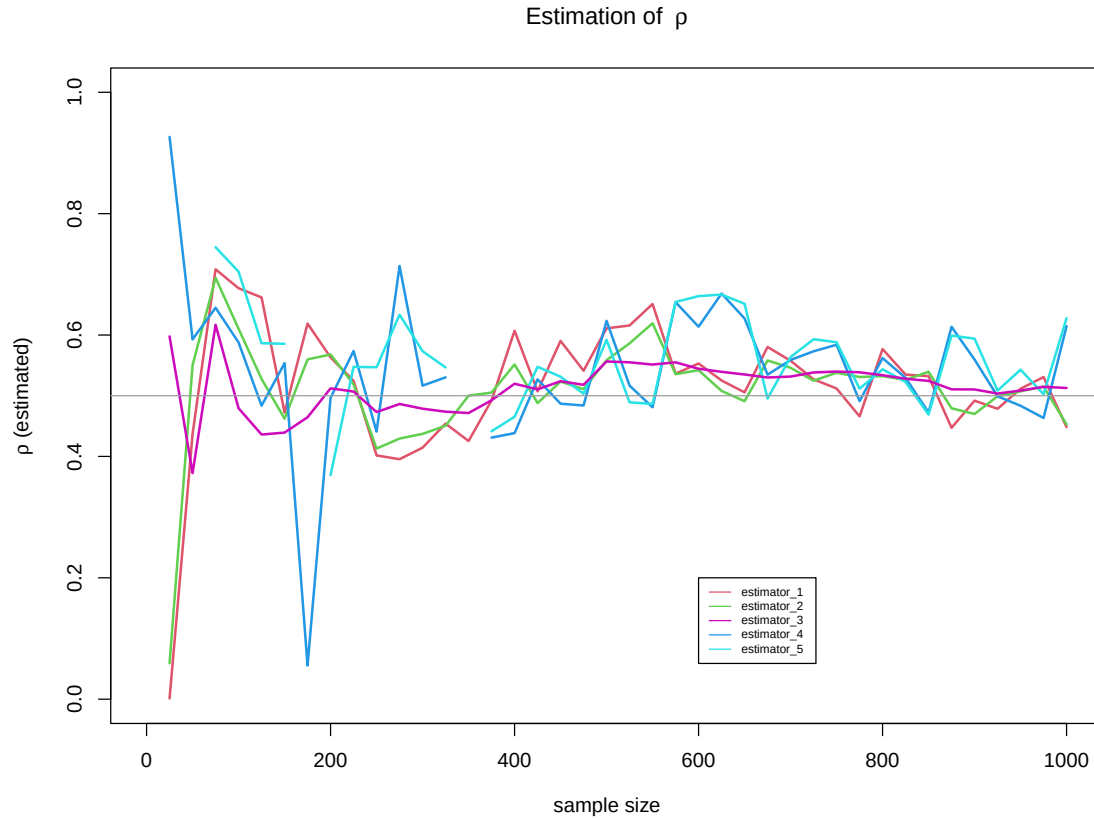


Figure 1: Comparison Between The Estimators

From the above plot, it is evident that some estimators (e.g., $estimator_1, estimator_2$) exhibit faster convergence toward the true parameter value as sample size increases,

indicating lower bias and variance. In contrast, other estimators (e.g., *estimator*₄, *estimator*₅) display greater variability, particularly at smaller sample sizes, which may be due to their sensitivity to outliers or instability under the heavy-tailed nature of the underlying distribution. The plot of *estimator*₄, *estimator*₅ are not continuous due to the occurrence of NA values.

Key Learninngs

Having fixed the true value of ρ at 0.5, the analysis of five different estimators reveals:

1. Overall Convergence: All estimators generally converge toward the true ρ value (0.5) as sample size increases, demonstrating the consistency property of the estimators.

2. Performance Ranking:

- `estimator_3` (middle line) shows the most stable performance across all sample sizes
- `estimator_2` and `estimator_4` exhibit moderate variability
- `estimator_1` (top line) shows significant overestimation
- `estimator_5` (bottom line) demonstrates consistent underestimation

3. Effects of Sample Size :

- Below $n=200$: Extreme volatility in all estimators
- $n=200-600$: Gradual stabilization begins
- Above $n=800$: Clear separation of estimator performance emerges

4. Optimal Estimator Characteristics:

- `estimator_3` maintains:
 - Minimal bias closest to true ρ
 - Lowest variance among all options
 - Fastest convergence rate
- Shows robustness even at small sample sizes ($n \leq 100$)

5. Practical Recommendations:

- For $n \leq 300$: All estimators remain unreliable
- For $n \geq 500$: `estimator_3` becomes clearly preferable
- `estimator_1` and `estimator_5` should be avoided due to systematic bias
- The trade-off between `estimator_2` and `estimator_4` may warrant further investigation

6. Theoretical Implications:

- The results confirm that the heavy-tailed nature affects estimation
- Traditional estimators (likely `estimator_1` and `estimator_5`) perform poorly
- Robust methods (likely `estimator_3`) handle the Cauchy distribution effectively

Resources & Tools

- **Books:**

- Vijay K. Rohatgi and A. K. Md. Ehsanes Saleh (1976). *An Introduction to Probability and Statistics*.
- Johnson et al. (1995). *Continuous Univariate Distributions, Volume 1*.
- Casella, G., and Berger, R. L. (2002). *Statistical Inference* (2nd ed.). Duxbury.

- **Software:** R (v4.3.2), Overleaf.

- **R Packages:** `psych`, `cauchypca`, `graphic`, `mass`, `writexl`.

Conclusion

In summary, the ratio-based statistic Z , derived from transformations of the Cauchy distribution, offers a theoretically sound alternative for estimating the correlation parameter ρ . This estimator leverages the robustness properties of heavy-tailed distributions and may provide insight into dependence structures under non-Gaussian settings. However, our simulation studies demonstrate that, despite the appeal of such non-classical approaches, the classical Pearson correlation coefficient retains strong empirical performance even when the underlying data deviate from normality.

Specifically, Pearson’s estimator remains relatively stable and accurate across a wide range of scenarios, including those with moderate to heavy-tailed marginal distributions. The comparative evaluation suggests that while the Cauchy-derived estimator can be useful in certain contexts—particularly for theoretical exploration or in the presence of extreme outliers—it is more sensitive to sample size and exhibits higher variance in small samples.

Future research could investigate hybrid estimators that combine the robustness of Cauchy-based methods with the efficiency of traditional estimators. Additionally, variance-stabilizing transformations or Bayesian shrinkage techniques could be explored to enhance the small-sample properties of the Cauchy-based estimators. Such developments may yield estimators that are not only theoretically robust but also practically reliable across a broader range of distributional assumptions.